Computational Companion to the Theory of Materialization: P0 Simulator, Calibration, and Phase Structure

Companion operational manuscript

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Abstract

This manuscript operationalizes the lowest crystalline and monatomic regime of the materialization program in computational form. It is a downstream companion to the core theory manuscript rather than a proof document for the theorem chain stated there.

The companion introduces a simulator-level translation map, a calibration layer, computational observables, admissible growth and escape moves, verdict rules, sweep protocols, and reproducibility requirements for the P0 regime. Laboratory validation is out of scope for the present document and is delegated to a separate laboratory companion.

The present pass is limited to the lowest regime $\mathcal{H}_0$ together with its P0 computational operationalization. No higher-level hierarchy closure, universal kinetics, or canonical calibration is claimed here.

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1 Protocol, Boundary, and Computational Status Policy

1.1 Purpose of the document

This manuscript translates the lowest regime of the core materialization theory into computationally manipulable objects. Its role is to define the simulator, the calibration layer, the observables, and the verdict logic needed for a reproducible P0 computational line.

1.2 Boundary of the computational manuscript

The present document is downstream to the core theory manuscript. It does not claim theorem closure of M-C1 or M-C2, universal physical kinetics, higher-hierarchy closure, or laboratory validation. It is restricted to the H0 and P0 regime: seeded crystalline growth in a computationally controlled monatomic setting.

Shared boundary clause. This companion is downstream to the core theory manuscript; it is not theorem evidence for the core manuscript; and it operationalizes only the lowest regime through a simulator-level calibration and verdict package.

1.3 Computational status labels

The computational companion uses a status system parallel to, but distinct from, the core theory manuscript:

- [CA] Computational Assumption.
- [CD] Computational Definition.
- [CP] Computational Protocol.
- [CR] Computational Result.
- [CF] Computational Failure Mode.
- [CT] Computational Target Claim.

2 Translation Map from Theory to Simulation

2.1 Theory objects carried into computation

The central task of the companion is to translate theory-level objects into simulation-level counterparts without collapsing the theory and the simulator into a single evidential layer.

Theory object	**Simulation object**	**Implementation choice**	**What is not claimed**
Σ_O^{mono}	T_{pkg}	Encoded target package with lattice class, thresholds, and persistence window	No claim that the package is a unique physical representation of the theory object
$R_{\mathrm{pre}}^{(a)}$	Reservoir state space	Finite precursor inventory and admissible occupancy states	No universal reservoir model
S_{seed}	Seed library	Enumerated nucleus templates and admissibility filters	No claim of exhaustive nucleation physics
Γ_{grow}	Growth engine	Ordered sequence of admissible local moves under a scoring rule	No claim of literal microscopic dynamics
Q_Λ	Order-score implementation	Lattice-order score computed from local and global pattern tests	No canonical order observable
$\lvert D(X)\rvert_O$	Defect surrogate	Weighted defect score from computational components	No observer-independent defect norm
$\mathcal{C}_O(X)$	Computational completion functional	Triplet of occupied-volume fraction, order score, and defect surrogate	No theorem-level equivalence to the theory functional
τ_O	Post-control test window	Finite persistence-testing horizon after control removal	No claim of universal metastability time

2.2 What remains surrogate-level

The simulator implements operational surrogates for order, defect, basin structure, escape barriers, and post-control persistence. These quantities are meant to support disciplined plausibility analysis, not to replace the theorem-level distinction between imported support, manuscript definitions, and target theorems.

2.3 Shared outcome ladder

The computational manuscript adopts the same high-level outcome ladder used across the companion package:

- fail before growth;
- growth fail;
- glass or amorphous capture;
- soft success; and
- hard success.

The computational companion assigns these classes by simulator-level observables and calibrated verdict rules rather than by theorem closure.

3 Computational Standing Assumptions

3.1 Geometric assumptions

Assumption 3.1 [CA]. Finite geometric regime. The computational regime uses a finite grid, a finite target window, and one target lattice class at a time. No continuum limit is claimed in the present computational pass.

In the current imported P0 implementation, the baseline geometry is a 16×16 discrete grid with a target window

$$T = \{4, \dots, 11\} \times \{4, \dots, 11\},$$

and an ideal even-even sublattice used as the monatomic crystalline target class. The resulting baseline completion volume is $v^* = 16$ ideal sites.

3.2 Composition assumptions

The present simulator is restricted to a monatomic or effectively one-species regime. Multi-species chemistry, mixed bonding classes, and compositional phase competition remain outside the current companion scope.

3.3 Dynamical assumptions

The simulator uses a finite admissible move set, local growth and escape operations, and a restricted family of post-growth escape paths. These constraints define an operational model class rather than a universal physical dynamics.

3.4 Observational assumptions

All diagnostics are observer-level diagnostics. The computational line does not invoke a hidden microscopic oracle that would bypass the coarse observables used in the completion functional and the verdict system.

4 Geometry, State Space, and Data Model

4.1 Fine grid and target window

Definition 4.1 [CD]. Computational target window. Let G denote the finite computational grid, $T \subseteq G$ the target window, and L_Λ the target lattice class. The simulator records growth only inside the finite window T while allowing external spill and escape events to be represented explicitly as off-target outcomes.

4.2 State representation

A simulator state may be represented either as an occupancy field on G or as a structured package

$$X = (I, S, E),$$

where I records ideal occupancy, S strained occupancy, and E external or contaminated occupancy. The document keeps the representation abstract so that later code imports can fix a concrete schema without changing the manuscript logic.

4.3 State classes

The baseline state classes are ideal occupied cells, strained occupied cells, and external contamination or off-target occupancy. These classes provide the minimal vocabulary needed to distinguish ordered growth, controlled nonideality, and invalid growth spill.

4.4 Run record schema

Definition 4.2 [CD]. Run record schema. A computational run record contains a configuration block, a growth-history block, a diagnostics block, a metastability block, and a verdict block. This schema is the minimal reproducibility contract for the P0 line.

The canonical imported run contract requires the outputs `calibration_scores.csv`, `capacity_sweep.csv`, `compatibility_sweep.csv`, `metastability_sweep.csv`, `verdict_atlas.csv`, `calibration_meta.json`, and `summary_report.md`, together with the associated plot files of the sweep campaign.

5 Computational Observables

5.1 Occupied-volume observable

The occupied-volume observable is written

$$V_{\mathrm{occ}}(X),$$

and records the fraction of the target window filled by admissible occupancy.

5.2 Order observable

The order observable is written

$$Q_\Lambda(X),$$

and measures how closely the realized state matches the target lattice class according to the current implementation of lattice-order diagnostics.

5.3 Defect components

The baseline defect components are

$$D_{\mathrm{coord}}, \qquad D_{\mathrm{hole}}, \qquad D_{\mathrm{off}}, \qquad D_{\mathrm{out}},$$

tracking coordination defects, hole defects, off-pattern occupancy, and out-of-window spill.

5.4 Defect surrogate

The defect surrogate is written

$$|D(X)|_O,$$

and is implemented as an observer-level weighted aggregation of the computational defect components.

5.5 Completion functional

Definition 5.1 [CD]. Computational completion functional. The computational completion functional is defined by

$$\mathcal{C}_O(X) = (V_{\mathrm{occ}}(X), Q_\Lambda(X), |D(X)|_O).$$

It is the operational counterpart of the theory-level completion functional for the lowest regime only.

5.6 Basin and persistence surrogates

Beyond the completion functional, the simulator records a basin tag, a local-minimum tag, a barrier surrogate, and a post-control persistence surrogate. These additional quantities support the computational hard/soft split without claiming a unique physical metastability theory.

6 Calibration Layer

6.1 Manual anchor states

Calibration begins with manually specified positive, negative, and borderline state families. These anchor states define the first coarse separation problem for the order, defect, and completion surrogates.

6.2 Mutation operators

The baseline mutation operators are deletion, strained insertion, external insertion, swap, and fragmentation. They are used to generate controlled perturbation families around the anchor states.

6.3 Mutation-generated calibration families

The calibration layer distinguishes near-good, borderline, and bad mutation families. The point is not to exhaust the full state space, but to force transparent separation tests for the computational observables.

The currently imported manual calibration library contains six good anchor states and eight bad anchor states. The good family includes `IdealFull`, `BoundaryVacancy`, `CornerVacancy`, `InteriorHole`, `TwoBoundaryDefects`, and `SlightlyIncompleteButOrdered`. The bad family includes dense amorphous, off-lattice dense, mixed crystal-strained, underfilled ideal, spill-outside, glass-like noisy, fragmented-islands, and borderline-bad states.

6.4 Calibration workflow

Protocol 6.1 [CP]. Calibration workflow. The frozen order of calibration is

$$K_\Lambda \to Q_\Lambda \to q^* \to |D| \to \mathcal{C}_O \to \text{basin} \to \text{escape family}.$$

The workflow moves from local pattern recognition to order scoring, threshold fixing, defect normalization, completion logic, basin tagging, and finally escape-family construction.

6.5 Calibration acceptance criteria

The current calibration acceptance tests are order separation, defect monotonicity, basin consistency, and barrier monotonicity under an enlarged admissible path family.

For the imported legacy P0 snapshot, the calibration report records

$$q^* = 0.6328,$$

with fourteen manual calibration states. The summary report also records best-good and best-bad values of the crystalline functional F_{cryst}, which are kept in the computational companion as release metadata rather than as theorem evidence.

6.6 Frozen calibration version

Calibration is versioned explicitly. Each frozen calibration snapshot must record the anchor-state library, mutation operators, threshold choices, and acceptance-test outcomes used in the active simulator release.

Calibration field	Representative symbol	Recorded meaning
Calibration identifier	κ_{P0}	Frozen version string for the active computational package
Order threshold	q^*	Active lower bound for the order score
Completion threshold	v^*	Active lower bound for occupied-volume completion
Defect budget	$\varepsilon_{\mathrm{comp}}$	Active upper bound for the defect surrogate
Persistence window	N_{hold}	Post-control hold horizon in simulator steps
Escape family tag	$\mathfrak{E}_{\mathrm{P0}}$	Active admissible escape-path family used for barrier tests

7 Admissible Move Set and Growth Engine

7.1 Growth moves

The baseline growth moves are `AttachIdeal`, `AttachStrained`, `Reject`, and `Stop`. These moves define the minimal growth vocabulary for the present monatomic regime.

In the current implementation, admissible ideal and strained attachment candidates are generated from an ideal frontier and a strained frontier. A move is admitted only when both the local mismatch score μ and the local defect increment Δ_{loc} satisfy the active bounds (μ^*, δ^*).

7.2 Escape moves

The baseline escape moves are `Detach`, `Slip`, and `OffTargetSpill`. They define the initial path family for post-growth and post-control escape analysis.

7.3 Growth front and local compatibility

The computational growth front is the set of admissible boundary sites at which a growth move may be applied without immediate rejection. Local compatibility is evaluated by a mismatch score together with the predicted defect jump of the candidate move.

7.4 Growth policy

The growth policy may be deterministic or controlled stochastic, but in either case it must expose its scoring rule, tie handling, and stop logic. This policy is part of the reproducibility contract rather than an invisible implementation detail.

The imported P0 growth engine is deterministic. It ranks admissible moves by a score built from the local mismatch penalty, the incremental defect penalty, the volume gain, and the hole-reduction term, then applies the first admissible move in the ordered list.

8 Defect Recurrence and Persistence Module

8.1 Defect recurrence surrogate

The simulator uses a computational counterpart of the ordered-growth defect recurrence. This counterpart is not claimed as a theorem-level derivation, but as a disciplined recurrence surrogate for tracking inherited defect load, injected disorder, and front mismatch.

8.2 Bounded-defect test

A growth history is defect-controlled only if the defect surrogate remains within the active bounded-defect protocol throughout the relevant growth window.

The current defect surrogate uses four components,

$$D_{\mathrm{coord}}, \quad D_{\mathrm{hole}}, \quad D_{\mathrm{off}}, \quad D_{\mathrm{out}},$$

with baseline weights $(1, 2, 3, 4)$ normalized by the target completion volume. This is the concrete current implementation of the observer-level defect surrogate for the imported P0 line.

8.3 Escape-family construction

Escape-family construction defines which post-growth paths are admissible when estimating whether the realized object remains trapped in the intended basin or collapses to a lower-quality regime.

8.4 Barrier surrogate

The barrier surrogate is written

$$\Delta^{\ddagger}_{\mathrm{esc}}(X),$$

and records an operationalized escape difficulty for the active admissible path family.

8.5 Hard and soft split

Protocol 8.1 [CP]. Computational hard and soft split. The simulator records soft success when completion is reached but the post-control persistence surrogate fails, and hard success when both completion and persistence pass under the active calibration package. This distinction is operational and does not by itself constitute theorem evidence for the core manuscript.

9 Computational Protocol

9.1 Static calibration protocol

This protocol fixes the anchor states, mutation catalog, thresholds, and active calibration version before any sweep campaign begins.

The canonical imported entry point is `software/src/p0_prototype/scripts/run_all.py`. A release-quality static calibration pass must therefore freeze the baseline configuration, the seed library, the calibration-state library, the mutation operators, and the order-negative family used in the calibration split before any sweep outputs are accepted.

9.2 Deterministic sanity runs

Deterministic sanity runs test that the simulator reproduces the intended qualitative distinctions between reservoir failure, growth failure, glass capture, soft success, and hard success under controlled settings.

9.3 Capacity sweep

The capacity sweep varies precursor sufficiency and target fill conditions to locate threshold behavior between underfilled growth and admissible completion.

In the current imported P0 release, the capacity sweep uses the ideal nucleus seed `S0_IdealNucleus` together with capacity ratios

$$0.5,\ 0.75,\ 0.9,\ 1.0,\ 1.1,\ 1.25,\ 1.5,\ 2.0.$$

9.4 Compatibility sweep

The compatibility sweep varies local compatibility parameters and lattice-alignment settings to map the transition between ordered growth, defect-heavy growth, and glass or amorphous capture.

The current imported sweep varies

$$\mu^* \in \{1.0, 1.5, 2.0, 2.5, 3.0, 4.0\}, \qquad \delta^* \in \{0.02, 0.05, 0.08, 0.12, 0.2, 0.3, 0.5\},$$

and uses the ideal nucleus `S0_IdealNucleus` together with the defective nucleus `S2_DefectiveNucleus` to expose both hard-success and glass-capture sectors.

9.5 Metastability sweep

The metastability sweep varies escape-family and post-control settings to expose the operational split between soft and hard success.

The current imported metastability sweep varies

$$\beta \in \{0.5, 1.0, 1.5, 2.0, 3.0, 5.0\}$$

while retaining the candidate state produced by the baseline seeded-growth line. Hard success is recorded only when the escape probability remains below δ_{rel} and the candidate is a local minimum with respect to the current escape family.

9.6 Joint phase atlas

The joint phase atlas records verdict classes over the combined sweep space together with replication rules, stop conditions, and acceptance criteria.

10 Verdict System

10.1 Verdict labels

Label	**Operational meaning**
`FAIL_RESERVOIR`	Growth fails before admissible ordered attachment becomes possible.
`FAIL_GROWTH`	Growth proceeds but never reaches the completion functional thresholds.
`GLASS_CAPTURE`	The run becomes trapped in a disordered metastable regime distinct from ordered crystalline completion.
`SOFT_SUCCESS`	Completion is reached, but post-control persistence does not pass.
`HARD_SUCCESS`	Completion and post-control persistence both pass under the active computational package.

10.2 Verdict rules

The verdict system maps the completion functional, the basin diagnostics, and the persistence surrogate into one normalized run label. Tie handling and ambiguous cases must be recorded explicitly rather than silently coerced.

10.3 Companion verdict crosswalk

High-level class	Computational label	Operational trigger
Fail before growth	`FAIL_RESERVOIR`	No admissible growth front or insufficient precursor support before ordered attachment begins
Growth fail	`FAIL_GROWTH`	Growth progresses but the completion functional thresholds are never jointly satisfied
Glass or amorphous capture	`GLASS_CAPTURE`	The run becomes trapped in a disordered metastable region distinct from the crystalline target class
Soft success	`SOFT_SUCCESS`	Completion passes while the post-control persistence surrogate fails
Hard success	`HARD_SUCCESS`	Completion and persistence both pass under the active calibration package

The current verdict implementation also records the growth termination mode. The imported modes are `RESERVOIR_FAIL`, `COMPLETE`, `CAPACITY_EXHAUSTED`, `NO_COMPATIBLE_STEP`, and `STEP_LIMIT_REACHED`. These modes are not identical to the verdict classes, but they are part of the run record needed to interpret verdict assignment.

10.4 Confidence and reproducibility

Verdict confidence depends on repeated runs, stability of classification under replication, and sensitivity to calibration or path-family perturbation. The verdict system is therefore inseparable from the reproducibility contract.

11 Computational Failure Modes

11.1 Calibration overlap

Failure Mode 11.1 [CF]. Calibration overlap. *Positive and negative calibration families cease to separate cleanly under the active order and defect observables. This failure invalidates claims of a stable verdict boundary.*

11.2 Frontier degeneracy

Frontier degeneracy occurs when the growth front ceases to distinguish admissible and nonadmissible attachment choices in a stable way.

11.3 False hard success

False hard success occurs when the persistence surrogate is too weak, too short, or too path-restricted to support the reported hard-success label.

11.4 Barrier hallucination

Barrier hallucination occurs when the escape barrier surrogate reports strong trapping but the admissible escape family is too narrow or too poorly calibrated to support that conclusion.

11.5 Verdict instability

Verdict instability occurs when small perturbations of calibration, replication seed, or path-family rules force repeated relabeling of the same run family.

11.6 Observer leakage

Observer leakage occurs when hidden implementation information bypasses the observer-level diagnostics meant to define the computational verdict.

12 Results

12.1 Anchor-state diagnostics

This section records the first stable separation of positive, negative, and borderline anchor states under the active order and defect observables.

The imported summary report already provides a first legacy snapshot of this section: the calibration package records $q^* = 0.6328$ over fourteen states, with a best-good F_{cryst} value of 0.0000 and a worst-good value of 5.6326.

The frozen calibration rows used by the current companion are listed below.

Label	**Family**	V	Q	d	F_{cryst}
IdealFull	good	16	1.0000	0.0000	0.0000
BoundaryVacancy	good	15	0.8789	0.1927	2.5163
CornerVacancy	good	15	0.8789	0.1667	2.3722
InteriorHole	good	15	0.8789	0.1979	2.4451
TwoBoundaryDefects	good	14	0.7656	0.3333	5.3444
SlightlyIncompleteButOrdered	good	14	0.7656	0.3854	5.6326
DenseAmorphous	bad	0	0.0000	2.0625	52.0406
OffLatticeDense	bad	0	0.0000	3.0000	56.7281
MixedCrystalStrained	bad	8	0.5000	3.0000	34.2281
UnderfilledIdeal	bad	8	0.2500	0.0729	20.3837
SpillOutside	bad	16	1.0000	1.0000	5.1500
GlassLikeNoisy	bad	6	0.2109	3.2500	42.7438
FragmentedIslands	bad	8	0.2500	0.1458	20.5684
BorderlineBad	bad	14	0.8235	0.9583	8.8653

12.2 Capacity threshold behavior

This section records how verdict classes change as precursor sufficiency and target fill capacity are varied.

The imported capacity sweep for `S0_IdealNucleus` reports three `FAIL_RESERVOIR` outcomes and five `HARD_SUCCESS` outcomes over the canonical ratio set.

The frozen capacity table for the imported release is:

Capacity ratio	**Reservoir units**	**Verdict**	V	Q	F_{cryst}
0.50	8	FAIL_RESERVOIR	4	0.0625	30.7281
0.75	11	FAIL_RESERVOIR	4	0.0625	30.7281
0.90	14	FAIL_RESERVOIR	4	0.0625	30.7281
1.00	15	HARD_SUCCESS	16	1.0000	0.0000
1.10	16	HARD_SUCCESS	16	1.0000	0.0000
1.25	19	HARD_SUCCESS	16	1.0000	0.0000
1.50	22	HARD_SUCCESS	16	1.0000	0.0000
2.00	30	HARD_SUCCESS	16	1.0000	0.0000

12.3 Compatibility phase structure

This section reports the computational phase structure separating growth failure, ordered growth, and glass capture under compatibility sweeps.

The imported compatibility sweep reports seventy-six FAIL_GROWTH outcomes, four HARD_SUCCESS outcomes, and four GLASS_CAPTURE outcomes across the current (μ^*, δ^*) grid.

All non-listed grid cells in the frozen release remain in FAIL_GROWTH. The only non-failure cells are:

Seed family	μ^*	δ^*	**Verdict**	Q	d
S0_IdealNucleus	3.0	0.3	HARD_SUCCESS	1.0000	0.0000
S0_IdealNucleus	3.0	0.5	HARD_SUCCESS	1.0000	0.0000
S0_IdealNucleus	4.0	0.3	HARD_SUCCESS	1.0000	0.0000
S0_IdealNucleus	4.0	0.5	HARD_SUCCESS	1.0000	0.0000
S2_DefectiveNucleus	3.0	0.3	GLASS_CAPTURE	0.9412	0.5250
S2_DefectiveNucleus	3.0	0.5	GLASS_CAPTURE	0.9412	0.5250
S2_DefectiveNucleus	4.0	0.3	GLASS_CAPTURE	0.9412	0.5250
S2_DefectiveNucleus	4.0	0.5	GLASS_CAPTURE	0.9412	0.5250

12.4 Order-defect separation plots

This section records plots and summary tables showing how order and defect surrogates separate or fail to separate the active calibration families.

12.5 Hard and soft split

This section records where completion passes but persistence fails, thereby marking the operational soft-success region.

The imported verdict atlas reports the joint counts

{FAIL_RESERVOIR : 36, FAIL_GROWTH : 36, SOFT_SUCCESS : 16, GLASS_CAPTURE : 12, HARD_SUCCESS : 8}.

These counts belong to the current imported legacy release and should be treated as baseline operational evidence for the P0 line rather than as theorem support.

The explicit metastability sweep underlying the soft/hard split is:

β	**Barrier**	τ_{meta}	p_{esc}	**Verdict**
0.5	2.3722	3.2743	0.9528	SOFT_SUCCESS
1.0	2.3722	10.7212	0.6065	SOFT_SUCCESS
1.5	2.3722	35.1046	0.2479	SOFT_SUCCESS
2.0	2.3722	114.9439	0.0833	SOFT_SUCCESS
3.0	2.3722	1232.3358	0.0081	HARD_SUCCESS
5.0	2.3722	141649.5192	0.0001	HARD_SUCCESS

12.6 Verdict atlas

This section collects the joint verdict atlas for the lowest regime together with replication notes and calibration version metadata.

The frozen atlas has a simple release pattern. For $\mu^* \in \{1.5, 2.5\}$ every tested run remains below the admissible growth regime: capacity ratio 0.9 yields `FAIL_RESERVOIR` and capacity ratios $1.1, 1.5$ yield `FAIL_GROWTH`, independently of $\beta \in \{1, 2, 4\}$. For $\mu^* \in \{3.0, 4.0\}$ the same capacity-ratio split at 0.9 remains `FAIL_RESERVOIR`, while the completion-capable runs bifurcate by seed quality and persistence strength: the ideal seed gives `SOFT_SUCCESS` at $\beta \in \{1, 2\}$ and `HARD_SUCCESS` at $\beta = 4$, whereas the defective seed gives `GLASS_CAPTURE` over the same completion-capable cells.

13 Interpretation Relative to the Core Theory

13.1 What the simulator supports

The computational companion supports disciplined plausibility analysis for the lowest regime by turning the core theory objects into explicit simulator objects with auditable verdict rules.

13.2 What remains surrogate-level only

Order scores, defect surrogates, escape barriers, and persistence tests remain simulator-level surrogates. They may support operational reasoning, but they do not by themselves close the theorem chain of the core manuscript.

13.3 Why this is not theorem evidence

This companion is not theorem evidence for the core theory manuscript. It belongs to downstream calibration and realizability work and must not be read as replacing the distinction between imported support, manuscript definitions, and target theorems.

13.4 Crosswalk to the laboratory companion

The computational companion provides the first operational crosswalk to the laboratory manuscript. Its target package, verdict ladder, and calibration identifiers are meant to prescreen apparatus-level runs rather than to bypass them.

In particular, simulator-level hard success is a release condition for laboratory escalation, not a substitute for laboratory persistence measurements.

14 What Is Not Claimed

The present computational companion does not claim a proof of M-C1 or M-C2, universal physical kinetics, canonical calibration, higher-level closure, or laboratory validation. It is a disciplined simulator manuscript for the lowest operational regime only.

A Symbol Table

This appendix collects the theory-to-simulation vocabulary, the active status labels, and the normalized run-record keys used throughout the computational companion.

Symbol or key	Normalized computational meaning
T_{pkg}	Encoded simulator target package derived from Σ_O^{mono}
$Q_\Lambda(X)$	Phase-anchored order score used in the imported P0 implementation
$d(X)$ or $\lvert D(X)\rvert_O$	Weighted defect surrogate normalized by the completion volume
F_{cryst}	Effective crystalline functional used for basin and calibration diagnostics
μ^*	Active local mismatch bound for admissible growth steps
δ^*	Active local defect-increment bound for admissible growth steps
δ_{rel}	Escape-probability threshold distinguishing soft and hard success
τ_O	Post-control test window used in metastability evaluation
`termination_mode`	Growth-engine stop code recorded in the canonical run report

B Calibration State Library

This appendix lists the positive, negative, and borderline anchor-state families together with their mutation-generated descendants.

Anchor family	Current count	Representative labels
Good family	6	`IdealFull`, `BoundaryVacancy`, `CornerVacancy`, `InteriorHole`
Bad family	8	`DenseAmorphous`, `OffLatticeDense`, `GlassLikeNoisy`, `FragmentedIslands`
Seed library	4	`S0_IdealNucleus`, `S1_LineSeed`, `S2_DefectiveNucleus`, `S3_BoundaryNucleus`

C Mutation Catalog

This appendix normalizes the permitted mutation operators, their parameter ranges, and their intended diagnostic purpose.

- `delete_ideal`: remove one ideal target site from the current state.
- `insert_strained`: add a strained occupancy site near the target pattern.
- `insert_external`: add an external spill site outside the target window.
- `swap_ideal_to_strained`: replace one ideal site by one strained site.
- `fragment_cluster`: remove a selected subset of ideal sites to produce disconnected islands.

D Parameter Tables

This appendix freezes the active threshold values, lattice-score parameters, defect weights, persistence-window settings, and sweep ranges for the current computational release.

Parameter family	Frozen value	Role in the current computational companion
Grid and target window	16×16, rows/cols $4 \dots 11$	Active lattice domain and central 8×8 target window
Completion threshold package	$v^* = 16$, $q^* = 0.6328125$	Lower bounds for admissible completion and order quality in the imported release
Compatibility package	$\mu^* = 2.5$, $\delta^* = 0.08$ baseline	Default local mismatch and defect-jump bounds outside dedicated sweeps
Defect weights	$(w_{\text{coord}}, w_{\text{hole}}, w_{\text{off}}, w_{\text{comp}}) = (1, 2, 3, 4)$	Aggregation weights for the four defect components
Persistence package	$\tau_O = 10$, $\delta_{\text{rel}} = 10^{-2}$	Post-control test window and hard-success escape threshold
Escape package	$\beta = 2.0$, $A_{\text{esc}} = 1.0$	Default post-growth escape-family parameters
Capacity sweep	$\{0.5, 0.75, 0.9, 1.0, 1.25, 1.5, 2.0\}$	Frozen reservoir-sufficiency ratios for the release sweep
Compatibility sweep	$\mu^* \in \{1.0, 1.5, 2.0, 2.5, 3.0, 4.0\}$, $\delta^* \in \{0.02, 0.05, 0.08, 0.12, 0.2, 0.3, 0.5\}$	Frozen compatibility grid
Metastability sweep	$\beta \in \{0.5, 1.0, 1.5, 2.0, 3.0, 5.0\}$	Frozen persistence-strength grid

E Frozen Result Tables

This appendix records the minimal release tables that define the imported P0 baseline independently of plots.

Sweep	Frozen pattern	Counts	Release reading
Capacity	Below ratio 1.0 fails, ratio ≥ 1.0 hard-succeeds	3 fail, 5 hard	First explicit reservoir threshold
Compatibility	Only $(\mu^*, \delta^*) \in \{(3, 0.3), (3, 0.5), (4, 0.3), (4, 0.5)\}$ escape	76 fail, 4 hard, 4 glass	Ordered-growth pocket depends on seed quality
	`FAIL_GROWTH`		
Metastability	Soft at $\beta \leq 2.0$, hard at $\beta \geq 3.0$	4 soft, 2 hard	Hard/soft split is persistence-driven
Verdict atlas	Joint counts over $(\mu^*, \beta, \rho_{\text{cap}})$ and seed family	36/36/16/12/8	Full lowest-regime operational ladder present

F Reproducibility and Code Provenance

This appendix records code provenance, run-environment metadata, calibration version identifiers, and reproducibility instructions for the active simulator package.

The active provenance bundle is anchored to the imported source tree `docs/Technology/programs/materia/sc` its canonical entry point `scripts/run_all.py`, and the required report products defined by the current P0 report contract.

G Run Book and Acceptance Checklist

This appendix contains the operational checklist for calibration, sweep execution, verdict assignment, and release gating of the computational manuscript.

1. Freeze the calibration identifier and threshold package before any sweep begins.
2. Record the active seed library, mutation catalog, and escape-family tag.
3. Execute sanity runs before parameter sweeps.
4. Record verdict counts together with ambiguous or tie-classified runs.
5. Block release if calibration overlap or verdict instability remains unresolved.